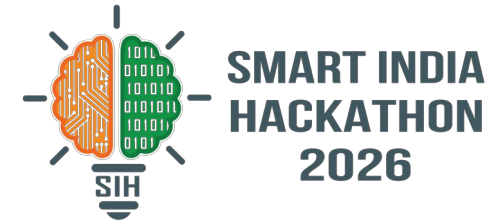


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – SIH26173
- **Problem Statement Title**- iTantra – Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for low bitrate links
- **Theme**- Smart Automation
- **PS Category**- Software
- **Team ID**- 188881
- **Team Name (Registered on portal)** Team Turtle



Website



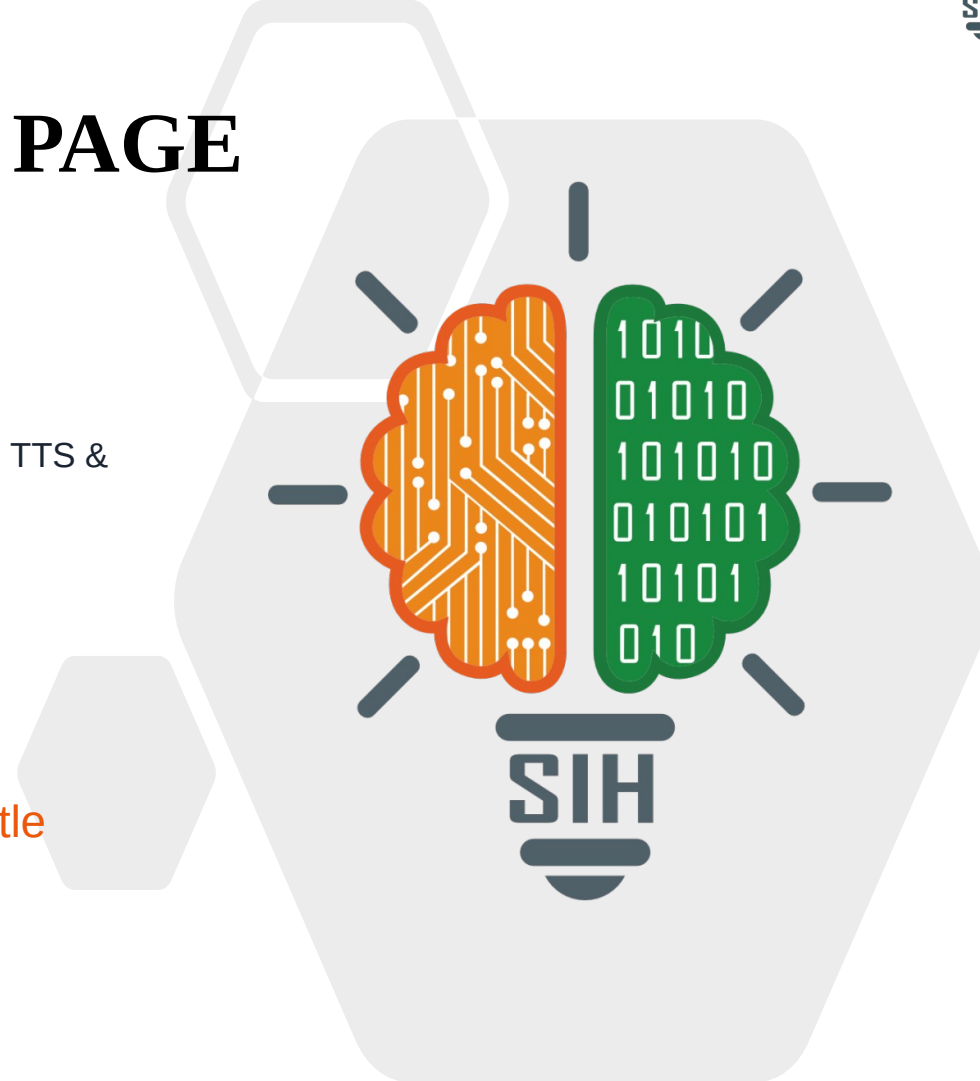
Demo film



Code

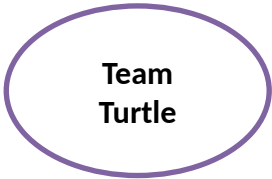


APK

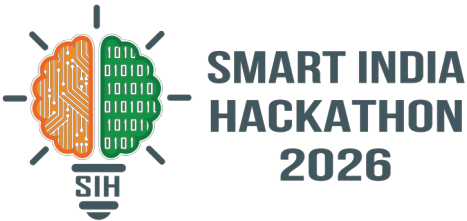


iTantra

Speak. Send text. Hear it back — offline.



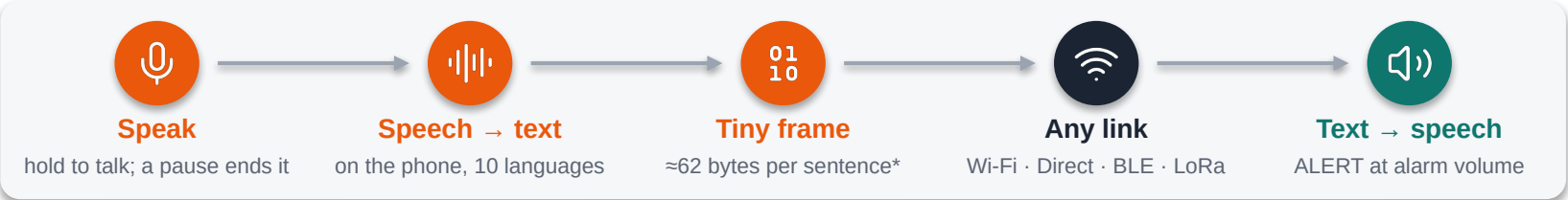
iTantra: Speak. Send Text. Hear It Back — Offline



Proposed Solution (Describe your Idea/Solution/Prototype)

Problem: when towers fail after a flood or cyclone, only thin links remain — Bluetooth, a crowded hotspot or LoRa radio at ~250 bps. Voice needs 256 kbps raw, ~12 kbps with a good codec.

Our idea: send the meaning, not the sound. Speech becomes text on the phone, only the text travels, and it is spoken again on arrival. **No tower, no internet, no cloud.**



Detailed explanation of the proposed solution

10 Indian languages

hi · bn · mr · te · ta · gu · kn · ml · or · en, speech in and out

ALERT that cannot be missed

Alarm stream, full volume, vibration (16/16 in test)

One-tap SOS with location

6 templates + GPS/NavIC, read aloud on arrival

Talk, call or captions

Push-to-talk, hands-free call, live captions

How it addresses the problem

Raw audio

256k

Opus voice

12k

iTantra

165

bits/s for one spoken sentence (log scale)

≈1,500× less data

than raw audio; fits LoRa's ~250 bps

62 B

per sentence vs 524 B as plain JSON

Output is still speech

— people who cannot read are included

Innovation and uniqueness of the solution

Speech itself is the codec

Text on the wire, voice rebuilt in the listener's language

One frame, any link

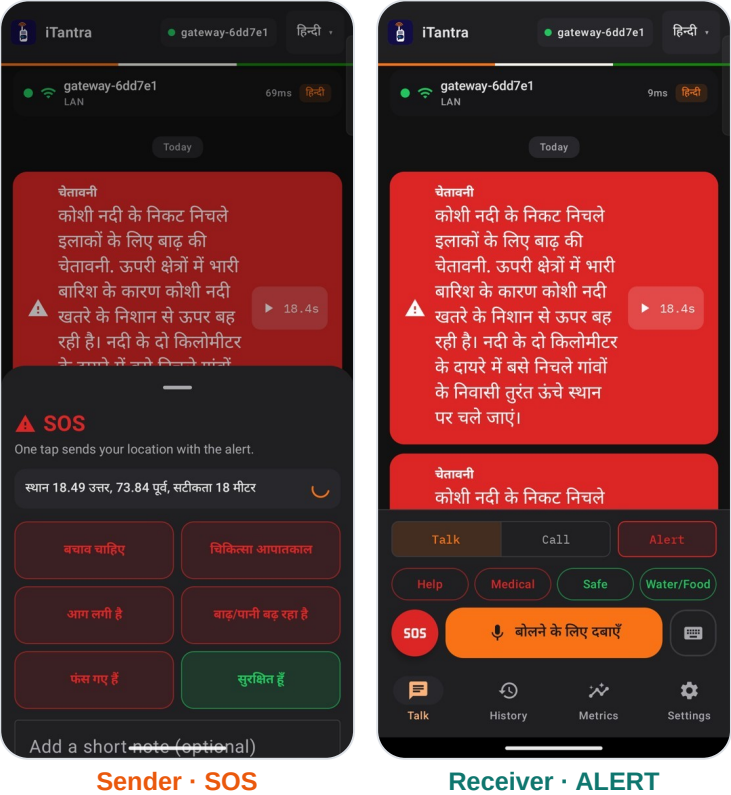
Wi-Fi → Wi-Fi Direct → BLE, no user action

Trusted alerts

Ed25519-signed ALERTs; encrypted after QR pairing

Adapts to the phone

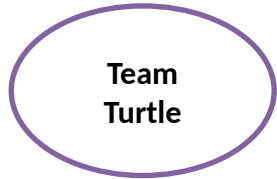
LITE profile switches on below 6 GB RAM



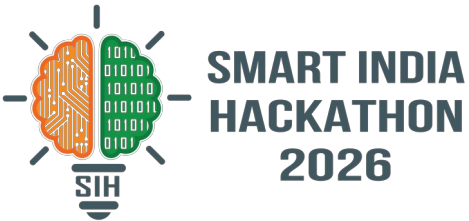
Watch the demo film · 1 min 55 s

Real screens from the running prototype (v0.1.0) on a vivo I2202, Snapdragon 870. [Try it: itantra-106.pages.dev](#)

*Median of 310 FLEURS sentences (unit test), before encryption and link headers. Bars assume a ~3 s sentence.



TECHNICAL APPROACH



Technologies to be used (e.g. programming languages, frameworks, hardware)

Android app
Kotlin 2.4 · Jetpack Compose · Android 8.0+, arm64

Voice detection
Silero VAD (0.6 MB): a pause ends the sentence

Speech-to-text
SraVaani-1.0 (IISc), int8 ONNX · sherpa-onnx

Text-to-speech
VITS Rasa (6 langs) · Indic-Mio (4 langs)

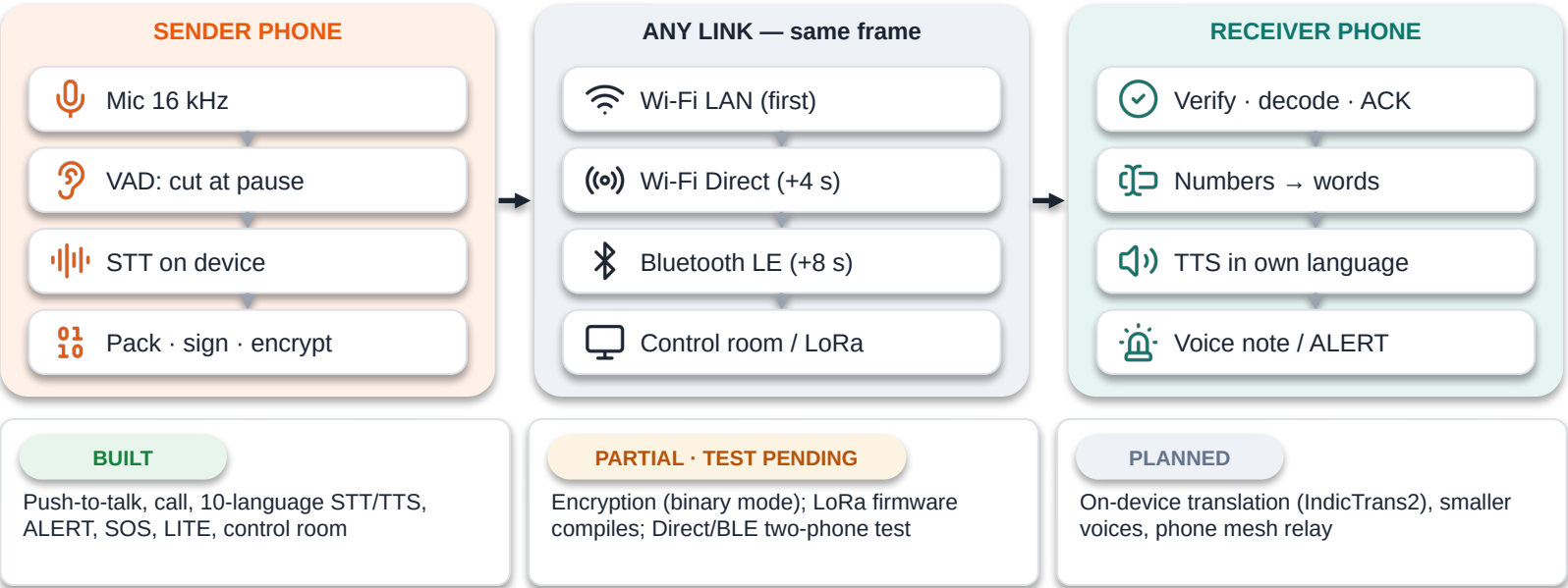
Links
Wi-Fi mDNS/TCP → Wi-Fi Direct → BLE GATT

Security
X25519 · ChaCha20-Poly1305 · Ed25519 · QR pairing

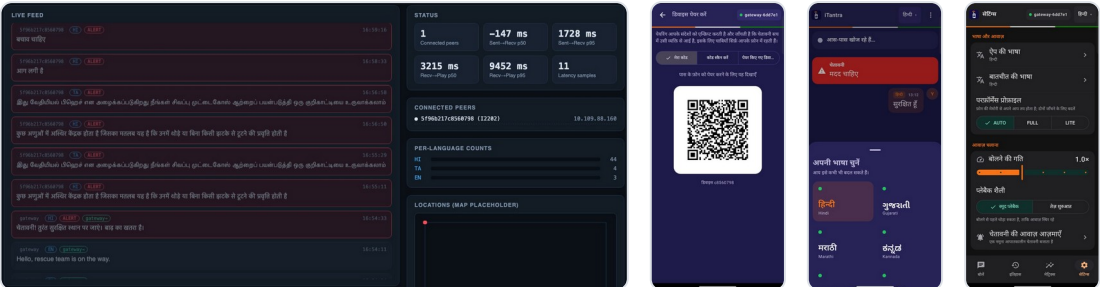
Radio hardware
ESP32-S3 + SX1262 LoRa, 865–867 MHz

Control room
Python gateway · SQLite · CAP 1.2 alerts

Methodology and process for implementation (Flow Charts/Images/ working prototype)



Working prototype — real screens



PC control room

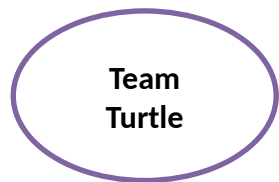
QR pairing

Language

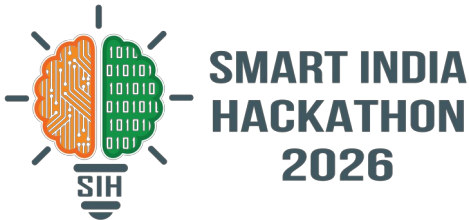
LITE profile

Prototype v0.1.0
APK 60 MB · Android 8.0+
Models 1.2 GB, installed once
Tested: Snapdragon 870 and 8 Gen 3

Reviewer install guide



FEASIBILITY AND VIABILITY



Analysis of the feasibility of the idea

Measured: 4 GB-class memory test, in-app logging

14x
faster than speech
STT real-time factor 0.07, 10/10 langs

0.6 s
speech end → text
median; range 0.42–0.70 s

765 MB
app RAM, idle
1.8 GB peak while speaking

0.3%
idle CPU
of one core, push-to-talk

Alive
app not killed
Android closed 115 other processes

Method: 12 GB vivo I2202 (Snapdragon 870) with RAM reserved so only ~1.7 GB stayed free, LITE profile. The chip was not slowed, so a budget phone will be slower. Also measured: 41 ms message → ACK over Wi-Fi; 0 audio underruns in 5 languages; word error rate 5.9% (preliminary, 10 clips).

Potential challenges and risks

Strategies for overcoming these challenges

Voice starts 3–6 s after arrival in LITE → Keep the listener's language loaded; small fast voices for hi/gu/or/en

1.14 GB of speech models → Per-language download on demand; installer checks SHA-256

Wi-Fi Direct / BLE: two-phone test pending → Two-phone field tests; Wi-Fi stays primary; LoRa bridge for range

Noise, accents, code-mixing → Alert-keyword boosting; 30-clip-per-language test; typed-text fallback

Viability

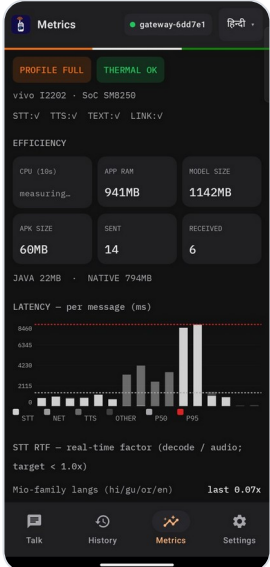
₹0 per message
No airtime, SIM, server or cloud fees

Phones people already own
Android 8.0+; LITE mode for less RAM

Low-cost range extender
ESP32-S3 + LoRa node ≈ ₹1,900–3,200

Open licences
MIT · Apache-2.0 · CC-BY-4.0 models

Fits India's alert flow
Control room reads CAP 1.2 alert files



In-app metrics (real screen)

Potential impact on the target audience



Disaster response teams

Coordinate rescue by voice when towers and internet are down



Coastal fishing communities

Cyclone warnings spoken in their own language, no data plan



Low-literacy users

An alert they can hear instead of read, in 10 languages



Remote field staff

A voice link for teams working where coverage is poor

≈1,500×

less data than raw audio*

1.03 s

ALERT arrival → spoken aloud (Wi-Fi)

10

Indian languages, speech in and out

Benefits of the solution (social, economic, environmental, etc.)



Social

Alerts spoken aloud in Indian languages, so people who cannot read are included. SOS carries location to rescuers.



Economic

No airtime, cloud or licence fees. Runs on phones people already own; a LoRa node costs ≈ ₹1,900–3,200.



Environmental

Fewer bytes per message means less radio time on air. Battery gain not yet measured.

Fits India's alert system: the control room reads CAP 1.2 alert files — the format behind [NDMA SACHET](#) — and re-broadcasts them as spoken alerts in each listener's language.

*62 B per ~3 s sentence (unit-test median) vs 256 kbps raw audio. No field pilot yet.

Photos, Wikimedia Commons: [NDRF – MHA](#), [GODL-India](#) · [fishermen – Saral Shots](#), CC0 · [woman on phone – Ammalu](#), CC BY-SA 3.0 · [rail workers – David Brossard](#), CC BY-SA 2.0

Details / Links of the reference and research work



Open models

[SraVaani-1.0 STT · ARTPARK, IISc](https://huggingface.co/ARTPARK-IISc/SraVaani-1.0)
huggingface.co/ARTPARK-IISc/SraVaani-1.0

[Indic-Mio TTS · SPRING Lab, IIT-M](https://huggingface.co/SPRINGLab/Indic-Mio)
huggingface.co/SPRINGLab/Indic-Mio

[VITS Rasa 13 TTS · AI4Bharat](https://huggingface.co/ai4bharat/vits_rasa_13)
huggingface.co/ai4bharat/vits_rasa_13

[Silero VAD](https://github.com/snakers4/silero-vad)
github.com/snakers4/silero-vad

[IndicTrans2 \(planned translation\) · AI4Bharat](https://github.com/AI4Bharat/IndicTrans2)
github.com/AI4Bharat/IndicTrans2



Runtimes and libraries

[sherpa-onnx](https://github.com/k2-fsa/sherpa-onnx)
github.com/k2-fsa/sherpa-onnx

[llama.cpp / ggml](https://github.com/ggml-org/llama.cpp)
github.com/ggml-org/llama.cpp

[mio-tts-cpp](https://github.com/mmnnga/mio-tts-cpp)
github.com/mmnnga/mio-tts-cpp

[Bouncy Castle \(encryption\)](https://www.bouncycastle.org)
www.bouncycastle.org

[RadioLib \(LoRa\)](https://github.com/jgromes/RadioLib)
github.com/jgromes/RadioLib



Standards and data

[OASIS CAP 1.2 alerts](https://docs.oasis-open.org/emergency/cap/v1.2/CAP-v1.2.html)
docs.oasis-open.org/emergency/cap/v1.2/CAP-v1.2.html

[NDMA SACHET](https://sachet.ndma.gov.in)
sachet.ndma.gov.in

[Google FLEURS \(STT test set\)](https://huggingface.co/datasets/google/fleurs)
huggingface.co/datasets/google/fleurs

[Codec2 \(bitrate baseline\)](https://github.com/drowe67/codec2)
github.com/drowe67/codec2

[Meshtastic \(LoRa mesh\)](https://meshtastic.org)
meshtastic.org



Android guides

[Wi-Fi Direct](https://developer.android.com/develop/connectivity/wifi/wifi-direct)
developer.android.com/develop/connectivity/wifi/wifi-direct

[Bluetooth and BLE](https://developer.android.com/develop/connectivity/bluetooth)
developer.android.com/develop/connectivity/bluetooth

[Audio focus \(ALERT\)](https://developer.android.com/media/optimize/audio-focus)
developer.android.com/media/optimize/audio-focus



Our work

[Source code](https://github.com/prasadxa/itantra)
github.com/prasadxa/itantra

[4 GB-class memory test](https://github.com/prasadxa/itantra/tree/main/tools/lowram)
github.com/prasadxa/itantra/tree/main/tools/lowram

[Model pack \(1.2 GB\)](https://r2.dev/models/v1/manifest.json)
r2.dev/models/v1/manifest.json

[Website and results](https://itantra-106.pages.dev)
itantra-106.pages.dev

[Demo film, 1 min 55 s](https://itantra-106.pages.dev/demo)
itantra-106.pages.dev/demo

[Reviewer install guide](https://itantra-106.pages.dev/#try)
itantra-106.pages.dev/#try



[Website](#)
overview · results



[Demo film](#)
1 min 55 s



[Code](#)
GitHub



[APK](#)
Android 8+